

A Remedy to Noise: Identifying and Adapting to Non-Linearity in Microcap Equities

Bennie Reynolds

Department of Economics, Clemson University

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Abstract

This study addresses whether predictable patterns exist in micro-cap excess returns, a segment often dismissed as idiosyncratic noise. In comparing linear models (IPCA), unstructured learners (XGBoost), and structured neural networks (Autoencoder), I find that linear models fail to sort the cross-section effectively, while the autoencoder achieves superior out-of-sample performance and decile monotonicity. These results prove that micro-cap returns are predictable when models account for non-linearities, such as threshold effects. Ultimately, a structured non-linear architecture successfully extracts systematic signal from a segment previously thought unforecastable.

1 Introduction

The micro-cap equity universe is historically been characterized by severe idiosyncratic volatility, leading many to dismiss its return variation as simply noise. This study questions if the micro-cap cross-section is purely governed by noise, and whether it contains predictable risk premia that traditional models fail to capture.

Empirical asset pricing models have historically been linear, and while proven effective in broad markets, they fail in the micro-capitalization segment ¹. Research focusing on micro-cap equities note how frequent delistings severely distort traditional linear inference (Eom and Park, 2023). In this study we compare modern linear and non-linear methods, in attempt to explain return variation within the micro-capitalization equity segment.

We find that modern models (IPCA, XGBoost, Neural Networks), all provide positive predictive power in the micro-cap segment when hyper-parameter tuned to maximize out-of-sample fit. Non-linear methods, particularly the neural network provides the strongest predictive power with low concern for overfitting. This not only proves that it is possible to model and explain some variation beyond general market exposure (beta) within micro-caps, but additionally that characteristic-factor dynamics in micro-caps are best fit non-linearly.

2 Literature Review

The cross-section of expected returns has traditionally been modeled through linear factor structures, from the Capital Asset Pricing Model (Sharpe, 1964) to the Fama-French five-factor model (Fama and French, 2015). While parsimonious, these models impose a static and homogeneous structure on factor loadings that may inadequately characterize heterogeneous market segments. This limitation is particularly relevant for micro-capitalization stocks. Micro-cap stocks occupy the second smallest segment of the public equity market ² and are characterized by high idiosyncratic volatility, low liquidity, and frequent delistings (Eom et al., 2023). Whether modern asset pricing methods can extract predictable variation from this segment remains an open question.

¹We define as the 20th-5th percentile of U.S. Equities by NYSE break-point

²The smallest segment being nano-cap stocks which are the 5th-0th percentile by NYSE break-point

2.1 From Linear Factors to Instrumented Models

Principal Component Analysis (PCA) extracts latent factors from return covariances without imposing pre-specified factor identities, modeling excess returns as:

$$R_{i,t} = \beta_i F_t + \varepsilon_{i,t} \quad (1)$$

Where β_i are factor loadings, F_t are factors, and $\varepsilon_{i,t}$ is the error term.

While this approach allows for firm-level loadings, these loadings remain static and estimated purely from returns, limiting both interpretability and out-of-sample generalization. Kelly, Pruitt, and Su (2019) introduced Instrumented Principal Component Analysis (IPCA) to address this constraint by modeling factor loadings as linear functions of observable firm characteristics:

$$r_{i,t} = \beta_{i,t-1}^\top F_t + \varepsilon_{i,t}, \quad \beta_{i,t-1} = \Gamma^\top Z_{i,t-1} \quad (2)$$

Where $Z_{i,t}$ is a vector of firm characteristics and Γ is a mapping from characteristics to factor loadings. This specification preserves PCA's variance maximization properties while restoring some economic context as instruments become time-variant risk exposures. In broad equity markets, IPCA explains over 35% of in-sample cross-sectional return variation, outperforming static factor models and PCA (Kelly et al., 2019).

Despite this success, IPCA assumes the characteristic-to-loading mapping is linear. This assumption may be violated in market segments dominated by liquidity and sparsity frictions.

2.2 Non-Linear Methods and the Autoencoder Architecture

Recent work has demonstrated that relaxing linearity improves return forecasting in broad equity markets. Gu, Kelly, and Xiu (2020) show that machine learning methods such as tree-based methods and neural networks, outperform linear benchmarks in cross-sectional prediction. Gu, Kelly, and Xiu (2021) subsequently propose an autoencoder architecture that preserves IPCA's factor structure while allowing non-linear characteristic-to-loading mappings:

$$\beta_{i,t} = g_\theta(Z_{i,t-1}) \quad (3)$$

Where $g_\theta(\cdot)$ is a feedforward neural network. This framework is a non-linear extension upon the latent factor decomposition framework of IPCA that preserves similar economic interpretability.

These methodological advances establish that non-linearity can improve asset pricing models. It remains unclear as to whether these improvements are uniform across market segments.

2.3 Theoretical Foundations for Non-Linearity in Micro-Caps

Three market frictions encompass the economic rationale for why characteristic-return relationships may be non-linear in micro-caps:

Limits to Arbitrage with Threshold Effects: Shleifer and Vishny (1997) argue that arbitrageurs deploy capital only when mispricing exceeds transaction costs and funding constraints. Micro-Caps average bid-ask spreads of 4-5% and the deployment of substantial capital may impact the market. Institutional Arbitrage therefore only becomes viable at extreme mispricings. These factors imply a threshold effect in which a characteristic signal must exceed a friction-dependent threshold before generating predictable returns. Linear models likely cannot best represent such discontinuities.

Institutional Ownership Constraints: Institutional investors face explicit minimum thresholds for market capitalization, share price, or liquidity (Gompers and Metrick, 2001). A stock trading at \$4.90 may be ineligible for institutional portfolios, while an otherwise identical firm at \$5.10 qualifies. These screens create step-function demand shifts that non-linear models should theoretically capture better than linear models.

Retail Attention Asymmetries: Retail investors dominate micro-cap trading due to institutional constraints Barber and Odean (2008) document that retail attention responds non-linearly to extreme returns, volume, or news coverage, generating disproportionate buying pressure in the tails of characteristic distributions. As a result, the predictive power of characteristics may exhibit non-monotonicity.

These frictions are endemic to micro-caps, but somewhat attenuated in large liquid stocks. If non-linearity arises from market frictions rather than mere statistical noise, non-linear methods should showcase stronger excess return prediction than linear methods.

2.4 Gap in the Literature

While Gu, Kelly, and Xiu (2021) establish that autoencoder architectures improve return prediction in the broad cross-section, their analysis does not test whether these gains are segment specific or examine the economic mechanisms underlying excess return non-linearity. Kelly, Pruitt, and Su (2019) demonstrate IPCA’s power in broad markets, but do not isolate micro-caps or test the linearity assumption in friction dominated segments.

This study addresses these gaps by conducting a systematic comparison of linear factor models (FF-5 and IPCA), an unstructured non-linear model (XGBoost), and a structured non-linear model (Autoencoder) within the micro-cap universe. In isolating micro-caps and decomposing model performance into market timing versus stock-selection components, we provide direct evidence on whether non-linearity is a general property of returns or a segment specific phenomenon driven by

identifiable economic frictions.

Furthermore, we test whether the optimal architecture preserves factor structure (autoencoder) or discards it entirely (tree-based methods), informing the methodological debate on integrating machine learning with economic theory.

3 Data

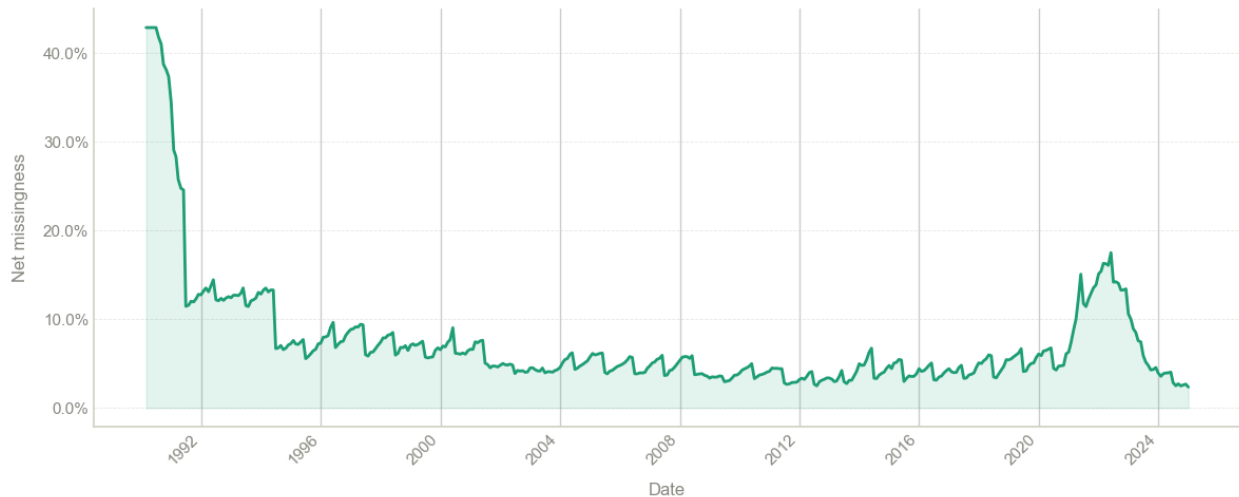
3.1 Data Overview and Sources

Our empirical analysis uses a panel of 11 characteristics spanning from January 1995 the last day of 2024. This sample period is selected to provide the largest time-frame for analysis as possible, whilst remaining faithful to the dynamics of the modern market. The dataset is constructed by merging data from the Center for Research in Security Prices (CRSP) with fundamental accounting data from Compustat. Using CUSIP bridging, we create a feature set that captures liquidity frictions, balance sheet health, and price momentum. *Table 1* provides a comprehensive taxonomy of these variables, and explains the time period by which variables are made available. Time-frame of availability is especially relevant for the accounting variables and associated derived metrics. One should make themselves familiar with the availability-schemes of each variable, to understand why characteristic loadings will need to be interpreted with nuance.

Mnemonic	Name	Source	Freq.	Definition	Unit
Market Data (CRSP)					
ret	Daily Return	CRSP	Monthly	Raw simple returns in-	Decimal
prc	Share Price	CRSP	Monthly	cluding distributions. Closing price or bid/ask	USD (\$)
shROUT	Shares Outstanding	CRSP	Monthly	average.	Thousands
mktcap	Market Cap	CRSP	Monthly	Total shares issued. Price $\times$ Shares Outstand-	Millions (USD)
vol	Trading Volume	CRSP	Monthly	ing / 1,000.	
bidask	Bid-Ask Spread	CRSP	Monthly	Number of shares traded during the period.	Shares
excess_ret	Monthly Excess Return	CRSP/FF	Monthly	Relative spread: (Ask - Bid) / Avg(Ask, Bid).	Percent
exchcd	Exchange Code	CRSP	Static	Risk-Free Rate. Listing exchange	Code
shrcd	Share Code	CRSP	Static	(1=NYSE, 2=AMEX, 3=NASDAQ). Type of security (10 or 11	Code
Fundamentals (Compustat)					
total_assets	Total Assets	Compustat	Annual	Total assets as reported	Millions (USD)
book_equity	Book Equity	Compustat	Annual	on the balance sheet. Stockholders' equity +	Millions (USD)
net_income	Net Income	Compustat	Annual	deferred taxes - preferred	
gross_profit	Gross Profit	Compustat	Annual	stock.	Millions (USD)
long_term_debt	Long-Term Debt	Compustat	Annual	Income before extraordi-	Millions (USD)
short_term_debt	Short-Term Debt	Compustat	Annual	nary items.	Millions (USD)
Derived Characteristics					
mktcap_lag	Mktcap Lag	Derived	Monthly	Total revenue minus cost	Millions (USD)
size	Log Market Cap	Derived	Monthly	of goods sold.	
momentum	Price Momentum	Derived	Monthly	Total debt obligations due	Millions (USD)
bm	Book-to-Market	Derived	Monthly	after one year.	Millions (USD)
roa	Return on Assets	Derived	Annual	Debt in current liabilities.	Millions (USD)
leverage	Market Leverage	Derived	Monthly		

3.2 Addressing Missingness and Delistings

Figure 1: Missingness over Time (%)



Microcap Equities exhibit a higher frequency of delisting and missingness than larger-cap equities. In the data cleaning process, measures were taken to minimize the impact of missingness on the validity of analysis. Due to lack of access to a CRSP-Compustat Linktable, fundamentals data is linked with 8-digit CUSIP bridging. Average market coverage throughout the sample is 84%. Average coverage from 2002-2020 is 89.38% which is remarkably close to 90% average matching rate of CRSP-Compustat linktable, matching rates from the years of 1990-2001 is 77.35% and matching rates in 2021 and 2022 being 71% and 64% respectively. These trends coincide with the Sarbanes-Oxley act in 2002 and lagged covid effects in 2021-2022. Smaller annual spikes suggest reporting patterns are responsible for the remainder of missingness.

If a delisting code (500-584) is encountered in the data pipeline, a -30% return penalty is applied to mitigate delisting bias (Schumway, 1997, p.340). To prevent look-ahead bias, fundamentals are lagged by 6-months and forward filled annually (Fama French, 1992, p.8). Micro-cap stocks are filtered using 20-5% NYSE breakpoints (Fama French, 1992, p.8).

3.3 Summary Statistics

Table 2: Summary Statistics

Variable	Obs	Mean	SD	Min	Median	Max
<i>Market Data (CRSP)</i>						
Monthly Return	505,086	0.008	0.158	-0.419	0.000	0.574
Share Price	505,064	13.688	10.456	1.000	10.990	57.500
Shares Outstanding	505,086	22,900	27,043	1,748	13,637	167,000
Market Cap (\$M)	505,064	200.054	152.015	19.745	154.476	672.083
Trading Volume	505,086	37,803	82,222	65.000	10,358	568,031
Bid-Ask Spread	505,086	0.048	0.028	0.004	0.042	0.155
Excess Return	505,086	0.006	0.159	-0.422	-0.002	0.570
<i>Fundamentals (Compustat)</i>						
Total Assets	404,676	518.793	814.478	5.109	202.554	4,622.773
Book Equity	404,571	123.362	141.615	-127.072	79.302	748.037
Net Income	403,904	-3.688	42.713	-245.860	4.094	91.666
Gross Profit	403,124	78.673	114.397	-67.603	43.523	698.588
Long-Term Debt	402,639	86.308	197.462	0.000	12.500	1,320.432
Short-Term Debt	403,187	23.058	63.631	0.000	1.676	422.221
<i>Derived Characteristics</i>						
Lagged Market Cap	505,086	198.106	147.212	20.505	154.733	635.893
Log Market Cap	505,086	4.995	0.806	3.021	5.042	6.455
Price Momentum	440,684	0.150	0.647	-0.813	0.035	3.243
Book-to-Market	404,571	0.634	0.597	-0.562	0.520	3.422
Return on Assets	403,904	-0.053	0.241	-1.295	0.012	0.289
Market Leverage	505,086	0.460	1.010	0.000	0.049	6.611

Note: After cleaning and linking the CRSP and Compustat datasets, the final sample consists of variable observations spanning 1990 to 2025. To mitigate the influence of outliers—particularly prevalent in the micro- cap segment—all continuous variables were winsorized at the 1% and 99% levels. The summary statistics in Table 2 reflect these adjustments, showing stabilized mean and standard deviation values compared to the raw returns and fundamentals data.

Table 2 presents summary statistics for the full 35-year sample and confirms the effectiveness of the data cleaning process. Mean and median weekly returns of 0.8% and 0% are consistent with a right-skewed return distribution. Price statistics follow CRSP negative price correction, with a minimum of \$1.00 consistent with exchange minimum listing requirements. The bid-ask spread minimum of 0.04% is plausible given the bid-ask midpoint formula, where negative values would be indicative of data quality issues. The mean bid-ask spread of 4.8% is also consistent with the

illiquidity characteristic to microcap firms.

Market Capitalization ranging from \$19.7-\$672 million confirms that the breakpoints captured the bulk of the micro-cap stock universe. With regard to fundamentals. Mean net income of -\$3.7 million with a positive median of \$4.1 million reflects skewness in the profitability among small firms, and mean ROA of -5.3% is consistent with the sample containing a meaningful share of early-stage and/or distressed firms. The leverage distribution with mean 4.9% and median 46% implies that microcaps within our sample predominately carry little debt, but a small subset are heavily leveraged. The trends observed within the summary statistics table are all theoretically consistent with characteristics of micro-capitalization stocks distributions and other literature analyzing small caps (Fama and French, 2008).

4 Methodology

4.1 Hypotheses and Evaluation Methodology I

Before conducting model-to-model comparisons, each model must first satisfy a baseline of predictability. we test the following hypothesis for each model:

- H_0 : The model's Predictive Out-of-Sample R^2 is less than or equal to zero ($R_{OOS}^2 \leq 0$).
- H_1 : The model's Predictive Out-of-Sample R^2 is strictly positive ($R_{OOS}^2 > 0$).

A positive R_{OOS}^2 indicates that the model has successfully identified a persistent economic signal that outperforms a naive forecast of zero excess returns. Models that fail this baseline are considered to have failed the fundamental task of asset pricing in the micro-cap universe.

4.2 Hypotheses and Evaluation Methodology II

For models that satisfy the baseline tests, we test three secondary hypotheses regarding the structural and statistical drivers of micro-cap excess returns:

- H_a : Selected models can provide predictive power differentiable from zero in micro-cap excess returns.
- H_b : Linear models outperform Non-Linear counterparts in predicting micro-cap excess returns.
- H_c : Unstructured models outperform Structured Non-Linear models in predicting excess returns.

These series of hypotheses will allow us to determine if our models can meaningfully learn the micro-cap return structure, and if the excess return structure is best characterized by linear or non-linear models.

4.3 Evaluation Metrics: Out-of Sample R-Squared and Noise Reshuffling

We utilize the monthly Out-of-Sample (OOS) Predictive R2 as the primary performance metric. This measures the proportionate reduction in the Mean Squared Prediction Error (MSPE) relative to the market average (Kelly et. al, 2019 and Gu, Kelly, Xiu, 2021):

$$\text{Mean-Based Predictive } R_{OOS}^2 = 1 - \frac{\sum_{(i,t) \in \text{test}} (r_{i,t+1} - \hat{E}_t[r_{i,t+1}])^2}{\sum_{(i,t) \in \text{test}} (r_{i,t+1} - \bar{r}_{t+1})^2} \quad (4)$$

Where $r_{i,t+1}$ is the realized excess return and $\hat{E}_t[r_{i,t+1}]$ is the forecast based on characteristics $Z_{i,t}$ available at time t . The predictive out-of-sample R2 will allow us to understand if a model provides a superior signal than the market average return. To test the amount of total realized movement in the market that can be explained by a model's structure, We secondarily use a zero-based mean predictive out-of-sample R^2 :

$$\text{Zero-Based Predictive } R_{OOS}^2 = 1 - \frac{\sum_{(i,t) \in \text{test}} (r_{i,t+1} - \hat{E}_t[r_{i,t+1}])^2}{\sum_{(i,t) \in \text{test}} r_{i,t+1}^2} \quad (5)$$

To ensure that R_{OOS}^2 results are not a product of overfitting to noise, we implement a noise shuffling protocol. We randomly permute the characteristic matrix Z to break the link between features and returns. A model is considered valid only if original performance is significantly higher than performance on shuffled data. To ensure out of sample integrity, the data set is partitioned chronologically: 60% training, 20% validation, and 20% test. This split prevents lookahead bias and p-hacking.

While Kelly, Pruitt, and Su (2019) emphasize results based on characteristic-managed portfolios, this study focuses on individual stock-level predictions. While effective for demonstrating factor risk premia, portfolioization tends to diversify away idiosyncratic volatility that characterized the micro-cap universe.

In utilizing a mean-based approach and strictly analyzing at the individual firm level (excluding FF5), this analysis subjects the IPCA, XGBoost, and Neural factor models to a more rigorous predictive hurdle. This approach ensures that the reported gains represent genuine informational advantage over an equal-weighted or value-weighted benchmark, rather than a mere reflection of broad systemic movements captured through aggregation.

4.4 Benchmark: The Fama-French 5-Factor Model

To establish a baseline, we evaluate the explanatory power of the Fama-French 5-Factor (FF5) model (Fama & French, 2015). The Fama-French 5 factor model specifies excess returns as a linear function of market, size, value, profitability, and investment factors:

$$R_{i,t} - R_{f,t} = \alpha_i + b_i(R_{m,t} - R_{f,t}) + s_iSMB_t + h_iHML_t + r_iRMW_t + c_iCMA_t + \varepsilon_{i,t} \quad (6)$$

This model provides a robust framework for broad-market equities, but suffers from two limitations when applied to micro-caps. The coefficients are assumed to be static over the estimation window. As high idiosyncratic volatility beyond beta exposure characterizes the micro-cap cross-section, static loadings cannot adequately describe an individual firm's risk profile (Kelly, Pruitt, Su, 2019). This modeling form will however theoretically provide a strong baseline for modeling broad market risk factors.

4.5 Instrumented Principal Component Analysis

Principal Component Analysis (PCA) is used to extract the latent factors that explain the maximum amount of common variance in returns. Given our panel of micro-cap returns $R_{i,t}$, we represent the return of asset i at time t as:

$$R_{i,t} = \alpha_i + \sum_{k=1}^K \beta_{i,k} F_{k,t} + \varepsilon_{i,t} \quad (7)$$

Where $F_{k,t}$ is the k -th latent factor, and $\beta_{i,k}$ is the static factor loading of asset i , and $\varepsilon_{i,t}$ is the idiosyncratic noise not explained by common factors.

IPCA extends PCA by modeling factor loadings as lagged firm characteristics. The dynamic loading is defined as $\beta_{i,t} = Z_{i,t-1}\Gamma$, where $z_{i,t-1}$ is a $1 \times L$ mapping matrix. IPCA becomes:

$$R_{i,t} = (Z_{i,t-1}\Gamma)F_t + \varepsilon_{i,t} \quad (8)$$

Following Kelly et al. (2019), I employ an Alternating Least Squares algorithm to minimize the objective function:

$$\min_{\Gamma, F} \sum_{i,t} (R_{i,t} - Z_{i,t-1}\Gamma F_t)^2 \quad (9)$$

The algorithm iteratively estimates factors for each month (F_t) and the mapping matrix (Γ) using Singular Value Decomposition. We apply the ALS estimator until stabilization $||\Gamma_j - \Gamma_{j-1}|| < 0.00001$. We define characteristic managed portfolios as: $x_{t+1} = \frac{Z_t^\top R_{t+1}}{N_t}$.

4.6 Extreme Gradient Boosting

To capture non-linearities and conditional dependencies in the micro-cap cross-section, we employ “eXtreme Gradient Boosting” (XGBoost). We define the predictive model for excess returns as an additive function of M trees:

$$E[R_{i,t+1}] = \sum_{m=1}^M f_m(Z_{i,t}) \quad (10)$$

Where $z_{i,t}$ is the vector of 11 rank standardized characteristics. The model is trained by minimizing a regularized objective function that balances predictive accuracy with model parsimony.

$$L(\phi) = \sum_i (R_{i,t+1} - \hat{R}_{i,t+1})^2 + \sum_m \Omega(f_m) \quad (11)$$

The complexity penalty ϕ includes both L_1 and L_2 regularization to prevent overfitting in the noisy micro-cap environment. We utilize hyperparameter tuning to identify the most effective tree depth, learning rate, and regularization coefficients.

4.7 Autoencoder

While XGBoost serves as a purely statistical predictive tool, we utilize a Neural Autoencoder to provide a non-linear alternative to Instrumented Principal Component Analysis. This model decomposes returns into a set of K latent factors and their corresponding characteristic-managed loadings:

$$R_{i,t+1} = \beta(Z_{i,t}; \theta) f_{t+1} + \varepsilon_{i,t+1} \quad (12)$$

The mapping $\beta(Z_{i,t}; \theta)$ is governed by a multi-layer perceptron (the “Beta-Network”). This allows the model to capture non-linear exposures to systematic risk. The Latent factors f_{t+1} are estimated via a bottleneck layer that compresses 11 characteristics into K factors. By jointly optimizing the network of parameters and latent factors, the Autoencoder generalizes the linear IPCA framework to accommodate the complex risk topographies inherent to micro-cap equities.

5 Empirical Analysis

5.1 Evaluation of Performance and Robustness

Table 3: Model Performance and Robustness:

Model Class	Predictive Performance		Robustness
	Zero-Based OOS Pred. R^2 (%)	Mean-Based OOS Pred. R^2 (%)	Noise R^2 (%)
FF5	-1.021466%	-1.567427%	-1.177448%
IPCA	0.1793%	0.0416%	-0.4430%
XGBoost	0.0861%	0.0747%	-0.0654%
Autoencoder	0.3027%	0.2113%	-0.0987%

Note: Noise R^2 is zero-based but characteristics are randomized on reshuffled characteristic-return pairs, confirming predictive collapse.

Our out-of-sample results illustrate progressive prediction improvement in model selection. The Fama-French 5-Factor model is unable to effectively explain excess return variance in the micro-cap segment. This finding reinforces the theoretical limitations of using portfolio break-points to model factor premiums when idiosyncratic volatility is high. FF5 model’s noise reshuffled R^2 is quite similar to its zero-based predictive R^2 . This verifies that FF5 is unable to meaningfully explain variance in the excess returns of micro-caps.

IPCA demonstrates positive predictive performance, with a 0.1793% zero-based predictive R^2 and a 0.0416% mean-based predictive R^2 . Whilst this explanatory power appears quite small Gu et. al’s 2021 paper, *Autoencoder Asset Pricing Models* found IPCA’s zero-based predictive R^2 to range between 0.10-0.31% in broad-equity markets. Our zero-based R^2 (0.1793%), closely aligns with their results, and our mean-based predictive R^2 (0.0416%) remains positive. We may assert that IPCA has used characteristic factors to explain excess-return variance beyond what can simply be explained by the market average exposures. IPCA’s noise reshuffled R^2 is negative and differentiable from zero, meaning that our model has likely not overfit to noise.

XGBoost exhibits a higher mean-based predictive R^2 than IPCA (0.0747%), with an inferior zero-based R^2 (0.0861%). We could gather that XGBoost’s decision trees are better at capturing relative over and under performers, whilst being less capable of explaining the level of returns. As XGBoost is a “local learner” that optimizes for squared error, whereas IPCA is designed to identify common market factors, these findings are not surprising, and somewhat reflect the classic bias/variance tradeoff. By assuming a linear relationship, IPCA is more rigid, allowing it to capture broad systemic signal more consistently (relative to XGBoost). XGBoost allows for

non-linear splits, which fit the cross-sectional mean better, but make its absolute predictions more volatile. XGBoost’s noise reshuffled R^2 is also negative and significantly worse than its other metrics, implying our concern for overfitting should be low, but higher than that of IPCA.

Autoencoder boasts the strongest results, with a zero-based predictive R^2 of 0.3027% and mean-based predictive R^2 of 0.2113%. The Autoencoder’s superior performance suggests that the complexity of the micro-cap segment is best navigated by a model that can both capture non-linear characteristic-loading mappings, and maintain a latent factor structure. It’s mean-based R^2 (more than double that of XGBoost), indicates that the deep-learning architecture is more effective at identifying relative “winners” and “losers”. It’s zero-based R^2 , suggests that it can also better capture systemic-factor variation, affirming that the model has not simply optimized the bias/variance tradeoff, but can explain more of the variation in micro-caps outright.

To contextualize these performance findings and ensure reproducibility, we will examine the parameterization that was found optimal by our hyperparameter testing. The following section details the hyperparameter configurations used in our best-fit models.

5.2 Model Parameterization

Table 4: Model Hyperparameters

Model	Category	Parameter (Setting)
IPCA	Structure	Latent Factors ($K = 2$)
	Regularization	Ridge Penalty ($\lambda = 1.00$)
	Estimation	Window (30-Month Expanding)
XGBoost	Structure	Max Depth (2), N Estimators (100)
	Learning	Rate ($\eta = 0.05$), Subsample (0.7)
	Regularization	$L1$ ($\alpha = 1.0$), $L2$ ($\lambda = 0.1$)
	Estimation	Window (Recursive Expanding)
Neural Factor	Structure	Latent Factors ($K = 5$), Hidden Layers (64, 32)
	Regularization	Dropout (0.10), Factor Ridge ($\lambda = 10^{-3}$)
	Complexity	Activation (ReLU), Optimizer (Adam)
	Training	Epochs (50), Learning Rate ($1e - 3$)
	Estimation	Window (60-Month Rolling)

The latent factor models are differentiated by optimal number of latent factors (K). The IPCA benchmark is constrained to 2 factors, contrasting the 5 factors of the Autoencoder. This asymmetry reflects the models’ differing capacity to handle high-dimensional interactions. IPCA fits best to a lower dimensional factor space in the micro-cap setting, whilst the Autoencoder’s activation function (ReLU) and dropout regularization (10%) presumably allow it to support a higher-dimensional

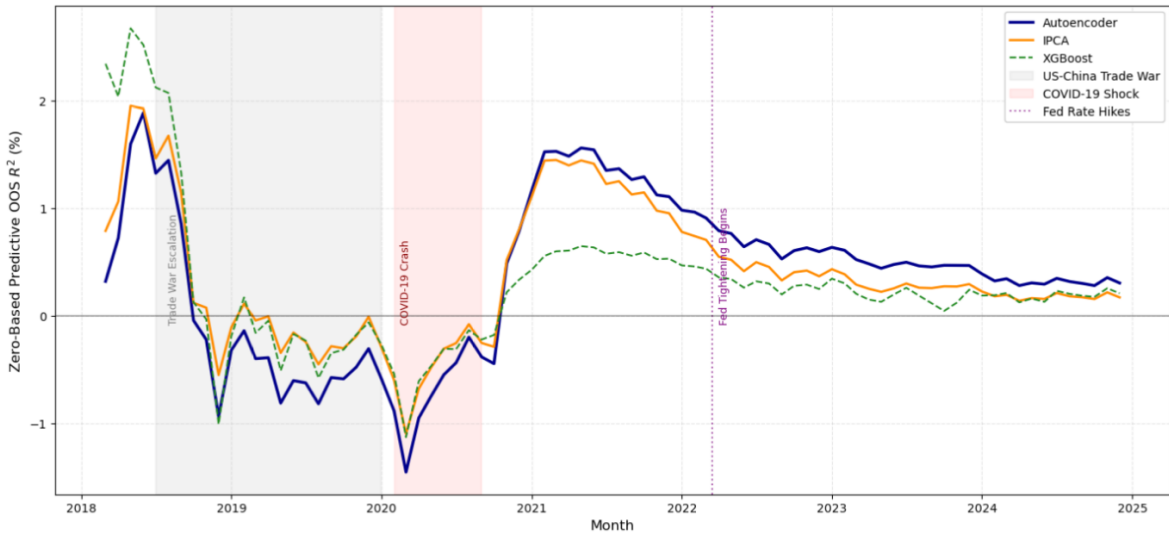
factor basis. IPCA employs a standard ridge penalty of 1.

XGBoost uses a combination of $L1$ (1.0) and $L2$ (0.1), although it should be noted that an $L1$ of 0.1 and $L2$ of 1.0 provides only marginally inferior out-of-sample predictive ability. XGBoost additionally employs a shallow tree depth (2) and a slow learning rate (0.05). The autoencoder integrates dropout during training, and a secondary ridge penalty ($\lambda=0.001$) specifically during the monthly factor estimation stage to ensure numerical stability.

The models are hyperparameter tested using varied temporal windows. The IPCA model best fits to a 30-month expanding window, and the autoencoder utilizes a 60-month rolling window. XGBoost follows a recursive expanding window approach, incorporating all available historical data as the model moves forward. The structural and regularization choices detailed in the model parameterization directly inform the performance trends observed in the rolling out-of-sample evaluation to follow.

5.3 Rolling Out-of-Sample R^2

Figure 2: Rolling OOS R^2 (2018-2025)



The performance wedge between the models highlights the tradeoff between the parsimonious $K=2$ IPCA and the higher capacity $K=5$ Autoencoder. The Autoencoder's ability to maintain a superior trajectory suggests the additional latent factors are successfully capturing non-linear risk dimensions not identifiable by a linear mapping. The stability of this lead validates the use of the 10% dropout rate and secondary ridge penalty as mechanisms for controlling model complexity.

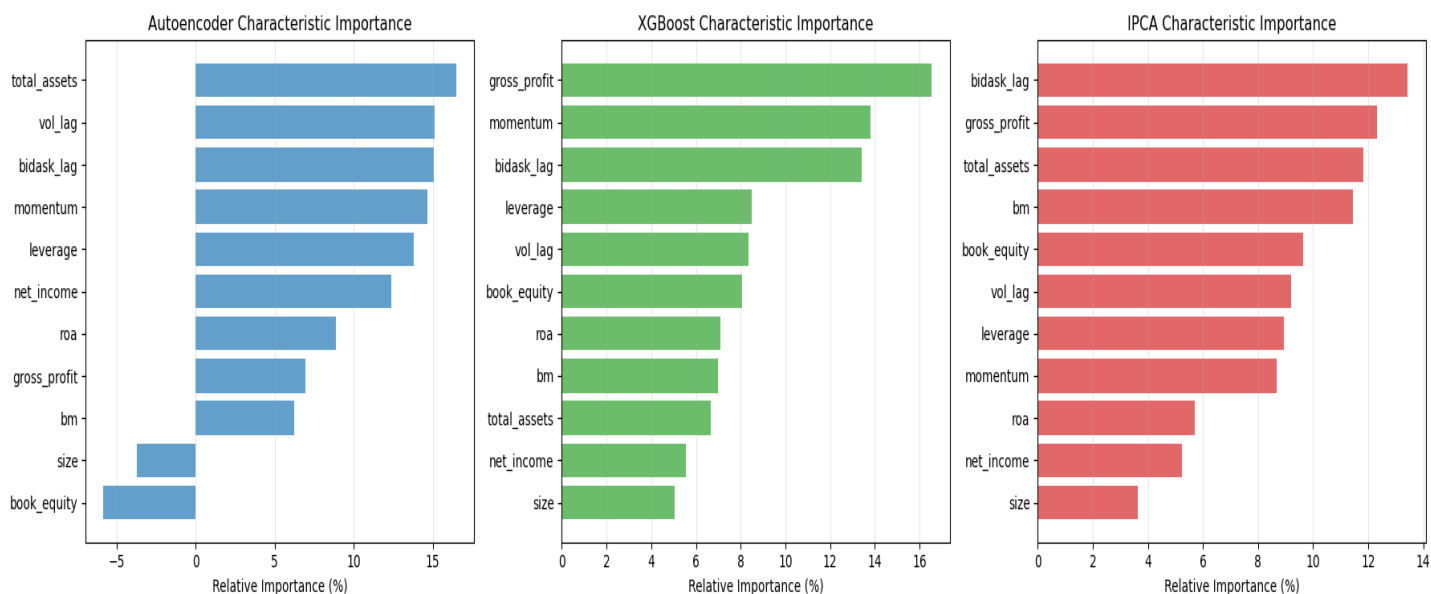
The temporal dynamics observed in the rolling out-of-sample graph highlight how regime factors impact model performance. The 2018-2019 window denotes a period of high model per-

formance. In 2019, all models tank in performance, which we can speculatively attribute to the US-China Trade War. The autoencoder underperforms other models in this period, possibly suggesting that in this case, model complexity leads to underperformance following structural breaks. Another performance crash can be seen in the overlaid period corresponding with COVID. After COVID, all models stabilize in performance, with the Autoencoder performing best. From this we can infer that the complexity of the autoencoder limits its' ability to perform in unstable regimes.

Having analyzed the performance and dynamics of model performance and specification, we now examine which specific firm characteristics drive the latent factors in each model. This may allow us to determine if the Autoencoder's performance is derived from traditional risk premia (e.g., Size, Value) or from non-linear interactions between liquidity and volatility indicators.

5.4 Characteristic Prioritization and Risk Drivers

Figure 3: Model Characteristic Importance



An analysis of the feature importances reveals divergence in how each model identifies the drivers of excess returns. IPCA prioritizes balance sheet scale and price friction characteristics. An interesting finding is that total assets are prioritized, while log-market cap (size) is negatively loaded. This suggests that the linear factor space identifies valuation discrepancies where asset-heavy firms are differentiated from their purely market-cap weighted peers. Combined with a high weighting on bid-ask spread, IPCA appears anchored in liquidity premia. XGBoost relies primarily on quality and trend indicators, with gross profit, momentum, and bid-ask spread forming its' dominant feature set. This configuration suggests the model prioritizes stocks with strong

fundamentals and established price trajectories. The inclusion of bid-ask spread implies the model identifies stocks with high relative visibility and active trading interest.

The Autoencoder demonstrates the most balanced distribution of importance across the characteristic set. Having bid-ask spread, gross profit, and total assets as its primary drivers, suggests a diverse emphasis on liquidity, quality, and relative scale. The autoencoder’s balanced information weighting suggests that its latent space is more diverse and complex, making identifying specific feature relations difficult when simply analyzing characteristic importance.

5.5 Partial Dependence Analysis

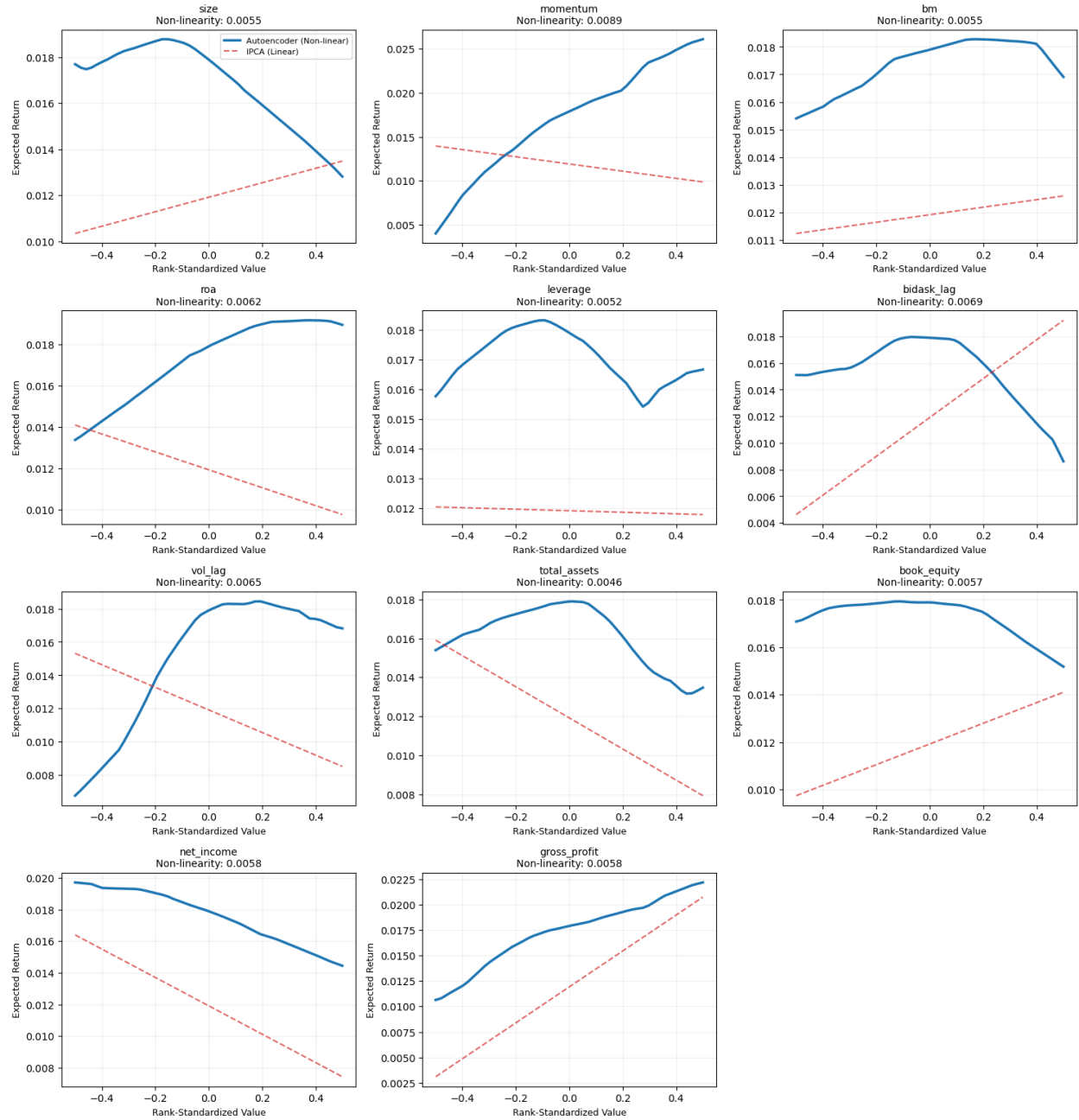
For brevity sake, I won’t analyze each plot *fig. 4* individually, but I will comment on trends easily observable. We can clearly identify that linear approximations are insufficient for most characteristics. Whilst IPCA is forced to draw constant slope lines on the relation of characteristics to excess returns, the autoencoder bends to fit diminishing return or saturation effects; these effects being particularly notable in bid-ask spread and total assets.

The non-linearity of the Autoencoder’s mappings provide superior performance, and better align with economic intuition. For instance, risk exposure should not logically scale linearly with bid-ask spread. In a real-world setting, some threshold might exist where spread transitions from a proxy for market visibility to a signal of heightened drawdown frequency or an uncertain outlook.

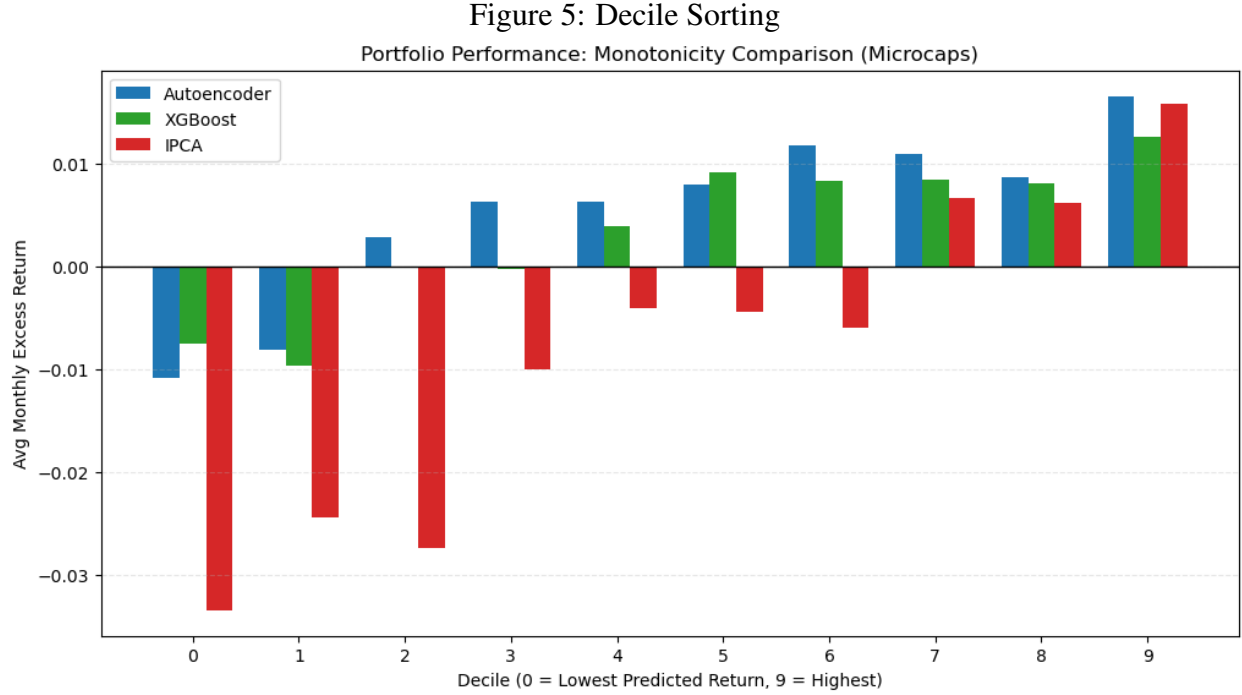
The partial dependence plots in concert with superior autoencoder performance, confirm the presence of non-linear sensitivities at the firm level. A final test of the empirical utility of our models, lie in their ability to identify actionable risk premiums. If a model has captured generalizable signals, we should observe an ordinal relationship between the model’s predicted returns and subsequent realized returns of the cross-section. To evaluate this, we move to an analysis of portfolio decile monotonicity, where we analyze the excess returns seen when we organize a portfolio sorted by expected return.

Figure 4: Partial Dependence Plots: Linear vs. Non-Linear Characteristic Mappings

Partial Dependence Plots: Non-Linear vs Linear Models



5.6 Analyzing Monotonicity of Portfolio Ordering



The Autoencoder exhibits a close to perfect ordinal progression from decile 1 to decile 10. IPCA's progression is far less monotonic, which is a clear struggle to effectively identify excess returns and ordinally organize until the decile 7, where it begins to predict excess returns more effectively. XGBoost also does not seem to be able to isolate monotonic progression, but identifies excess returns seemingly better than IPCA in its portfolio.

This mapping demonstrates the necessity for linear mapping when it comes to progressively organizing excess returns, but does suggest the use of IPCA for a long-only top decile portfolio may see similar return to a similar use case of the autoencoder.

6 Discussion

6.1 Addressing Hypotheses

In our R^2 figures and portfolio sorting, we prove that IPCA, XGBoost, and the Autoencoder all exhibit positive predictive power differentiable from zero. We reject H_a , and posit micro-cap excess return patterns can be non-trivially explained.

The empirical evidence provides a clear rejection of H_b . The out-of-sample performance disparity between the Autoencoder and IPCA confirms that linear models inadequately characterize

the micro-cap segment. As established by Shleifer and Vishny (1997), the presence of significant transaction costs in micro-caps creates “limits to arbitrage”. IPCA’s failure to produce positive excess returns until the 7th decile suggests that linear models poorly resolve “threshold effects”. The Autoencoder’s superior R^2 metrics and monotonic sorting indicate that non-linear mappings best capture the discrete demand shifts driven by institutional screens and liquidity constraints.

The comparison between XGBoost and the Autoencoder addresses whether factor structure allows for stronger performance. While XGBoost captures non-linear relationships, its failure to maintain clear monotonicity in the middle deciles support the preservation of a factor structure. We can infer that in constraining a non-linear mapping through the autoencoder bottleneck, the Autoencoder filters out statistical noise that XGBoost mistakes for signal. This confirms that integrating economic theory (latent factor structure) is superior to purely statistical learning, at least in the micro-cap cross-section sample we have analyzed. We reject H_c .

6.2 Limitations

The findings within this research may not generalize to other equity markets. Although Gu et. al (2021), established the presence of non-linearity in broad-equity markets, asset classes where arbitrage is more efficient may not follow these findings. The non-linearities we document are plausibly specific to the market frictions endemic to micro-caps.

While we heuristically label economic mechanisms, the mappings provided by our models are statistical constructs optimized for prediction. Inference regarding risk premia remains speculative. With more granular data and larger sample spans, emergent phenomena may surface. Further analysis with less restriction as it pertains to financial statement release and additional characteristic factors, would likely lead to more conclusive findings.

6.3 Extensions

Regime Dependent Models: The time-variation in autoencoder predictive power suggests that factor premia are regime-dependent. Future work could incorporate: VIX, credit spreads, liquidity conditions.

Structural Identification of Latent Factors: While our factors are statistically extracted, connecting them to theoretical risk sources (arbitrage capital, retail sentiment, funding liquidity) would strengthen economic interpretation. This could involve instrumental variable approaches or equilibrium restrictions.

Advancing Architecture: Recent advances in attention mechanisms (Transformers) and graph neural networks could capture cross-sectional dependencies (e.g., industry co-movement, supply chain linkages) that neither IPCA nor standard autoencoders exploit.

7 Conclusion

This study conducted a systematic comparison of asset pricing architectures within the micro-cap universe, a segment historically characterized by its defiance of traditional linear factor models. Our findings lead to following conclusions.

While linear models may suffice for large cap equities (Kelly et. al, 2019), the micro-cap cross-section is governed by systems that rewards non-linear mappings. As the Autoencoder model (first established by Gu et. al, 2021) is a non-linear extension upon IPCA’s linear factor mapping, the improvement in performance is not simply a product of a more complex model, but proof that non-linear factor mappings better explain variance in the micro-cap universe.

The autoencoder’s ability to maintain strict monotonicity over XGBoost, proves that an economically structured and non-linear approach is the optimal architecture for explaining micro-cap return variance. This approach bridges the gap between economic theory and machine learning. The fact threshold and saturation effects are observable in our partial dependence plots, provides empirical substance to the theories of “Limits to Arbitrage” and “Institutional Constraints”.

By isolating the micro-cap linear segment this research demonstrates that return variation in micro-caps can be non-trivially identified, and that the evolution from linear to non-linear mappings is necessary to best explain the underlying economic mechanisms within.

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